

BleedSense: Cross-Dataset Robustness for Surgical Blood Segmentation

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Abstract—Automatic hemostasis management in robotic surgery requires reliable segmentation of blood in the operative field. Models trained on a single surgical dataset, however, tend to generalize poorly to other datasets with different visual characteristics. This work studies the issue on two public datasets, HemoSet (robotic surgery on a porcine model) and the dataset of Rabbani et al. (gynecological laparoscopy on humans), confirming an asymmetric domain gap already observed in previous work from the research group. Six families of strategies for reducing this gap are then compared systematically: aggressive data augmentation, appearance/domain adaptation (Reinhard, FDA), multi-domain joint training (with and without balancing), ensembling of multiple architectures, an illumination-invariant input channel, feature-level style mixing (MixStyle), and a test-time BatchNorm recalibration. Across three standard architectures (UNet, UNet++, DeepLabV3+, resnet34 encoder), aggressive data augmentation is the most effective strategy: it improves cross-dataset robustness in both directions with no appreciable in-domain cost, and combined with an ensemble of the three architectures it yields the best overall result. The reasons behind the weaker performance of the remaining strategies are also analyzed.

Index Terms—semantic segmentation, domain generalization, cross-dataset robustness, surgical blood detection, data augmentation, domain adaptation

I. INTRODUCTION AND MOTIVATION

Automating hemostasis management in robotic surgery – identifying and controlling bleeding during the procedure – requires, as a first step, reliable segmentation of blood in the endoscopic field of view. This project is part of the TRAMIS research initiative, focused on real-time bleeding detection in robotic surgery. Real, annotated TRAMIS surgical data were not yet available at the time of this work, so the study was conducted on two public reference datasets in the domain, used as proxies:

- **HemoSet** [1]: induced bleeding during teleoperated robotic surgery on a porcine model, 962 image-mask pairs, 10 subjects.
- **Rabbani et al.** [2]: gynecological laparoscopy on human patients, 751 annotated images.

Previous work from the research group [9] had observed an *asymmetric* performance drop when a model trained on one of the two datasets is evaluated on the other: HemoSet→Rabbani shows a strong general performance drop, while Rabbani→HemoSet shows a marked increase in false positives (non-hemorrhagic regions segmented as blood).

Project goal: to quantify this asymmetry rigorously and reproducibly, and to systematically study which strategies actually increase the cross-dataset robustness of segmentation models, comparing multiple experimental conditions without the constraint of converging on a single solution.

II. RELATED WORK

Segmentation models for surgical and medical imaging are well known to generalize poorly across acquisition setups, instrumentation, and patient populations, a domain gap that is especially acute for a visually heterogeneous, rare target such as intraoperative bleeding. HemoSet [1] and the dataset of Rabbani et al. [2] are, to our knowledge, the two public benchmarks for this specific task, respectively covering robotic surgery on a porcine model and human gynecological laparoscopy, each evaluated with its own protocol (IoU/F1/Hausdorff Distance, and IoU/F-Score). A first quantification of the asymmetric cross-dataset gap between the two, which is the problem addressed in this work, was reported in an unpublished prior student project [9]; it is used here only as a starting reference point for that observation.

Two broad, architecture-agnostic families of techniques have been proposed to close such domain gaps without redesigning the network. The first acts at the appearance level, aligning low-level image statistics between source and target domains: explicit color transfer [4], or exchanging low-frequency Fourier amplitude components while preserving phase (FDA) [3]. The second acts at the feature level inside the network: MixStyle [5] mixes encoder feature statistics across batch samples during training, while test-time BatchNorm recalibration (AdaBN) [6] instead adapts normalization statistics to the target domain at inference, with no retraining required. Su et al. [7] show, separately, that a sufficiently varied data augmentation policy can act on its own as an effective, architecture-free single-source domain generalization strategy, a finding that the results reported here corroborate. A more structural alternative is explicit style-content disentanglement, pursued by Teevno et al. [8] for endoscopic segmentation and, in the same prior student project mentioned above [9], through a boundary-aware Transformer architecture; both require dedicated architectural components rather than only training-time interventions.

This work addresses a distinct question from all of the above: rather than proposing a new architecture, we system-

atically test which of the simpler, purely training-time interventions – augmentation, appearance transfer, joint training, and ensembling – actually improve cross-dataset robustness for this task, and why some succeed while others do not.

III. APPROACH

A. Datasets and Splits

HemoSet consists of 962 image-mask pairs from 10 subjects (pigs); we use a Stratified Group K-Fold split into 5 folds, with the subject as the group, to avoid the same animal appearing in both training and validation within the same fold. Rabbani consists of 751 images/masks with a fixed 70/15/15 split (train/val/test, seed 42), reused identically across all phases of the project.

B. Architectures and Training Protocol

We use three standard architectures via `segmentation_models_pytorch`, all with the same `resnet34` encoder for a fair comparison: UNet, UNet++, and DeepLabV3+. Training uses a combined Dice+BCE loss, the AdamW optimizer (lr 10^{-4}), a `ReduceLROnPlateau` scheduler, batch size 8, 256×256 images, and the full dataset for each split. Early stopping on validation Dice is used (patience 8, cap of 40 epochs); it was verified empirically, across all 87 Phase 1–4 runs, that this budget was sufficient for convergence (only 2/87 runs reached the cap, both with a residual improvement below 0.01 Dice). The exact same configuration is used for every condition compared, to isolate the effect of the single variable under study.

C. Metrics and Evaluation Protocol

We report Dice, IoU, Precision, Recall, F1, and HD95 (95th-percentile Hausdorff Distance). This set fully covers the metrics used in the original papers of both datasets, with HD95 preferred over the maximum HD because it is less sensitive to single-pixel outliers. A model trained on HemoSet is evaluated in-domain on the validation fold and cross-dataset on the fixed Rabbani test set; a model trained on Rabbani is evaluated in-domain on its own fixed test set and cross-dataset on the whole of HemoSet (never seen during training). This makes the two datasets’ results statistically asymmetric: HemoSet figures are mean $\pm$ standard deviation over 5 folds, while Rabbani (which has no natural grouping variable to fold over) contributes a single run; Rabbani numbers should therefore be read with correspondingly more caution.

IV. RESULTS

A. Baseline and Domain Gap Asymmetry

The three architectures, trained separately on the two datasets with minimal augmentation, confirm and quantify the asymmetry: for HemoSet→Rabbani, mean Dice drops by about 70% relative ($0.745 \rightarrow 0.217$), with HD95 rising from ~ 29 px to ~ 130 px, driven by a collapse in recall ($0.76 \rightarrow 0.19$) – the model becomes too conservative on the target domain and misses most of the actual blood. For Rabbani→HemoSet

the drop is more contained ($0.684 \rightarrow 0.585$), but precision collapses instead ($0.74 \rightarrow 0.50$) while recall stays high, consistent with the original observation about false positives. The three architectures behave very similarly under every condition, and none of them resolves the domain gap on its own. These baseline numbers are quantitatively consistent with the zero-shot estimates reported in [9] (Dice 0.29 and 0.56 respectively) despite the different splits and hyperparameters used, which is taken here as a sanity check on the measurements rather than a direct comparison.

B. Data Augmentation, Appearance Adaptation, Joint Training

An “aggressive” augmentation policy (hue/saturation, gamma, blur, noise, JPEG compression, via `Albumentations`) improves cross-dataset robustness across all three architectures in both directions, at almost no in-domain cost (HemoSet→Rabbani: $0.217 \rightarrow 0.321$, +48% relative; Rabbani→HemoSet: $0.585 \rightarrow 0.601$). Reinhard color transfer [4] and FDA [3], applied using unlabeled target-domain images, help the more critical direction as well (HemoSet→Rabbani: $0.217 \rightarrow 0.263$ and $\rightarrow 0.242$ respectively), but at a non-negligible in-domain cost (Rabbani in-domain: $0.684 \rightarrow 0.611$ with Reinhard), and neither method beats aggressive augmentation overall. A possible explanation is that adaptation, unlike augmentation, is always active during training rather than applied with some probability per image: the model never sees the “clean” appearance of its own source domain, and ends up split between two styles instead of properly consolidating one with variations around it. Fig. 1 shows two independently-sampled draws of the aggressive policy on representative training images, illustrating what is actually being varied beyond what the metrics alone convey.

Training on the union of HemoSet and Rabbani (joint training, with separate held-out test sets) was not a solution: there is no systematic benefit on HemoSet, and a systematic drop on Rabbani ($0.684 \rightarrow 0.636$) caused by the size imbalance between the two datasets in the combined training set ($\sim 59\%$ HemoSet vs $\sim 41\%$ Rabbani). A domain-balanced sampling scheme (`WeightedRandomSampler`, $\sim 50/50$ per batch) recovers only about half of the drop ($0.636 \rightarrow 0.656$), which is not enough to make joint training competitive with the other strategies.

C. Ensemble of the Three Architectures

As an additional methodological choice of our own, the three architectures’ predictions are ensembled at zero additional training cost, simply by averaging the sigmoid probabilities of the already-trained aggressive-augmentation checkpoints. The result beats the best single model on all four tested conditions, with modest but systematic, exception-free gains (e.g., HemoSet→Rabbani: 0.337 best single model $\rightarrow$ 0.343 ensemble; Rabbani→HemoSet: $0.615 \rightarrow 0.623$), consistent with the standard ensembling rationale: errors made independently by architecturally different models tend to disagree

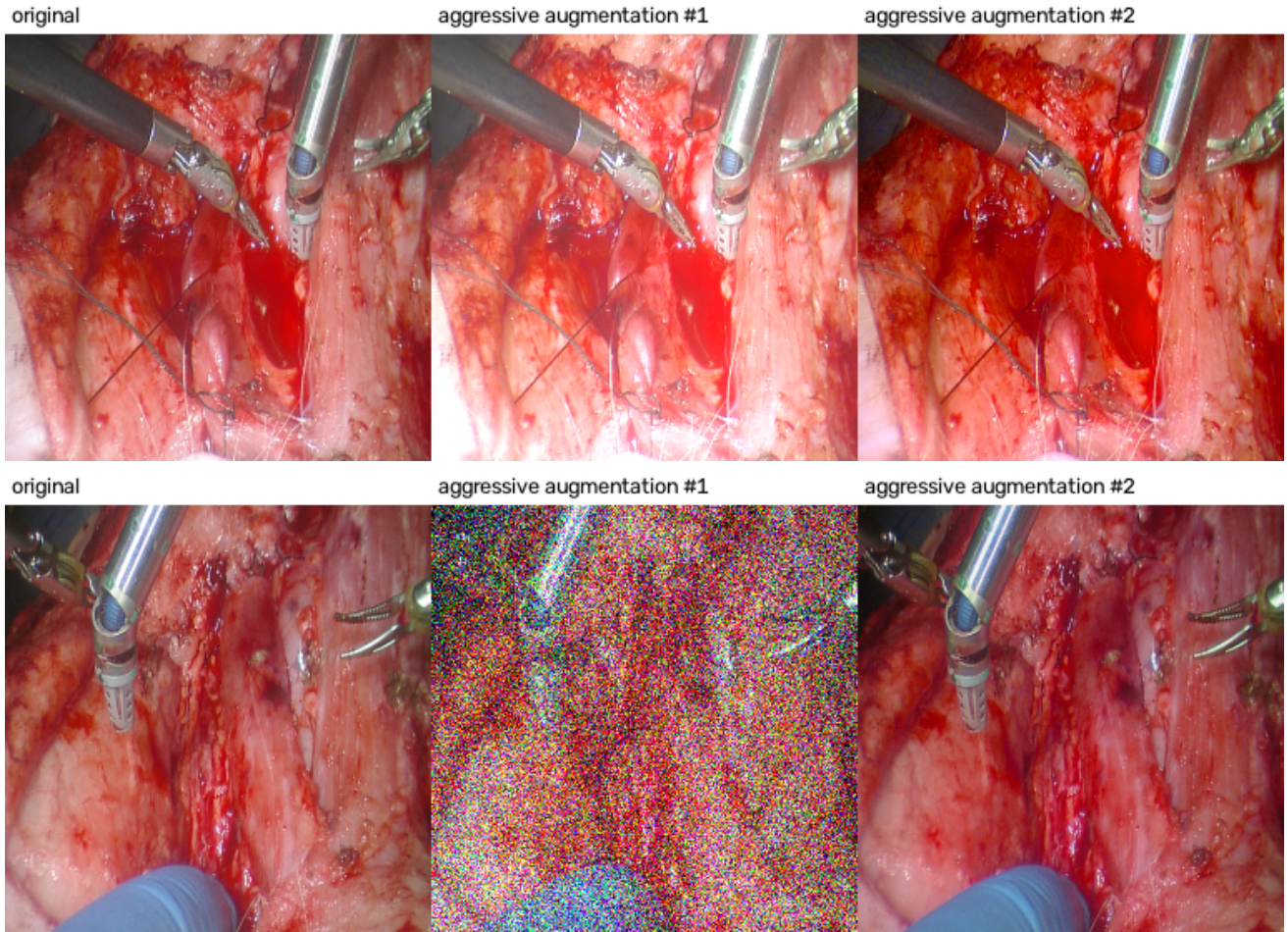


Fig. 1. Two representative HemoSet training images (left) alongside two independently-sampled draws of the aggressive augmentation policy (hue/saturation, gamma, blur, noise, JPEG compression, plus geometric flips) applied to each. The stochastic, per-sample nature of the policy is visible: the two draws for the same image differ from each other.

and partially cancel out when their probabilities are averaged, while the shared, correct signal is reinforced.

A practical caveat is worth noting given this project’s real-time motivation (Section I): the ensemble triples inference cost (three forward passes instead of one) relative to a single model. This is an acceptable trade-off for the offline evaluation performed here, but a deployed real-time bleeding-detection system would need to weigh the modest gains reported above against the added latency, or fall back to aggressive augmentation alone – already the single most effective and cheapest strategy tested – if strict real-time constraints apply.

D. Comparative Summary

Table I reports the mean over the three architectures for each strategy; full per-architecture results are in the supplementary materials.

Recommended configuration: aggressive augmentation in training + ensemble of the three architectures at inference – the only combination that improves cross-dataset robustness in both directions without penalizing in-domain performance.

E. Qualitative Results

Figures 3 and 4 show representative examples for each cross-dataset direction, comparing the ground-truth mask against the predictions of the source-only baseline, the aggressive-augmentation model, and the ensemble, overlaid on the original image. The baseline under-segmentation on HemoSet→Rabbani (missed blood regions) and the over-segmentation on Rabbani→HemoSet (false-positive regions) are visually apparent, and both are progressively mitigated by aggressive augmentation and further by the ensemble.

V. DISCUSSION

A. Why Augmentation Beats Adaptation and Joint Training

The common factor behind the two less effective strategies – always-active appearance adaptation and imbalanced joint training – appears to be the lack of a mechanism that preserves the model’s exposure to the “clean” distribution of its own source domain during training. A light, probabilistic intervention on visual variety outperforms a more targeted but always-active intervention on style, and it also outperforms simply

TABLE I
MEAN DICE AND HD95 (PIXELS) BY STRATEGY AND EVALUATION CONDITION, AVERAGED OVER THE 3 ARCHITECTURES.

Strategy	HemoSet (in-dom.)		Rabbani (in-dom.)		H→R (cross)		R→H (cross)	
	Dice	HD95	Dice	HD95	Dice	HD95	Dice	HD95
Baseline (source-only)	0.745	28.9	0.684	48.0	0.217	129.9	0.585	58.7
Aggressive augmentation	0.747	28.3	0.681	41.4	0.321	93.9	0.601	54.3
Adaptation (Reinhard)	0.719	32.2	0.611	70.2	0.263	101.0	0.577	54.9
Adaptation (FDA)	0.720	30.5	0.675	47.8	0.242	107.9	0.550	66.1
Natural joint training	0.742	28.6	0.636	58.5	n/a*	n/a*	n/a*	n/a*
Balanced joint training	0.739	31.3	0.656	55.2	n/a*	n/a*	n/a*	n/a*
Ensemble (3 arch.)	0.770	24.6	0.706	39.6	0.343	91.5	0.623	49.6

*joint training is evaluated in-domain on both test sets, so there is no comparable cross-dataset condition for this row.

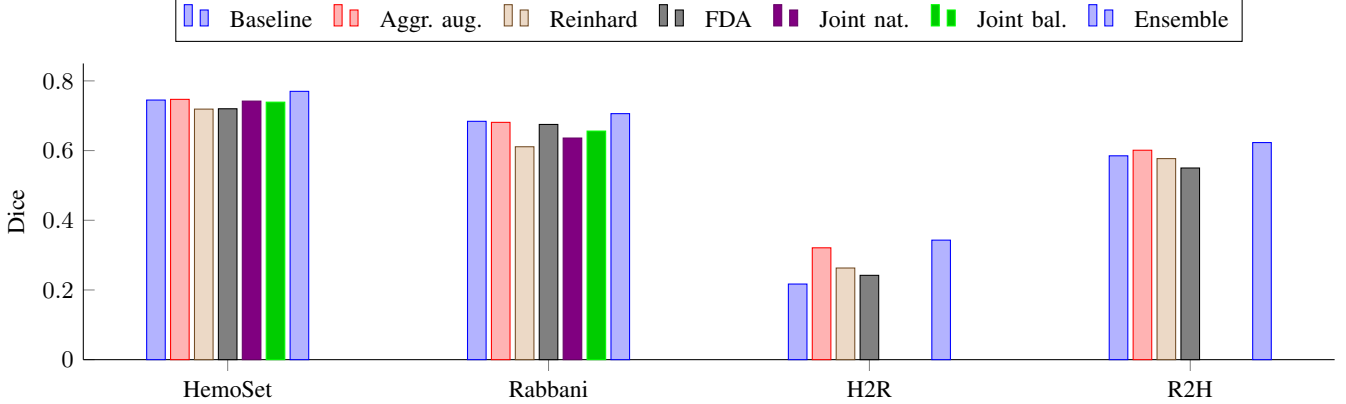


Fig. 2. Mean Dice by strategy and evaluation condition (averaged over the 3 architectures). HemoSet/Rabbani = in-domain; H2R = HemoSet→Rabbani (cross); R2H = Rabbani→HemoSet (cross). Joint training bars are omitted in the cross-dataset columns (not applicable, see Table I).

mixing the data together. This is visually corroborated by the qualitative examples in Figs. 3–4: aggressive augmentation and the ensemble progressively recover the recall lost by the source-only baseline on HemoSet→Rabbani, and reduce, without fully eliminating, the over-segmentation observed in the opposite direction.

B. Three Further Extensions (Negative but Informative Outcome)

After identifying the recommended configuration, we attempted three additional original extensions:

- 1) **“Blood-index” input channel** (ratio $R/(R+G+B)$, in theory less sensitive to camera/illumination differences): this slightly improves HemoSet→Rabbani ($0.321 \rightarrow 0.331$) but markedly worsens Rabbani→HemoSet ($0.601 \rightarrow 0.526$), since the index, computed on raw pixels, ends up carrying over the white-balance differences between the datasets instead of ignoring them, becoming a shortcut feature specific to the source domain.
- 2) **MixStyle** [5]: mixes the encoder’s feature-map statistics between samples in the same batch. The result is a diffuse but small degradation in both cross-dataset directions ($0.217 \rightarrow 0.203$; $0.585 \rightarrow 0.561$); a plausible reason is that it is applied too early in the encoder, and that training batches remain single-domain in this setting, which limits the actual style diversity injected.

- 3) **Test-time BatchNorm recalibration** (AdaBN [6]): recalibrates BatchNorm statistics on unlabeled target-domain images at inference time, with no additional training. It shifts the precision/recall trade-off in opposite directions in the two cross-dataset conditions, but net Dice does not improve in either one ($0.321 \rightarrow 0.318$; $0.601 \rightarrow 0.570$).

None of the three beats the recommended configuration, although the pattern is informative: three very different mechanisms fail in the same way, either by introducing a shortcut specific to the source domain or by disturbing useful information without replacing it with something more transferable. This reinforces the conclusion that aggressive augmentation combined with ensembling is not a local optimum that can be easily improved on with light interventions, at least for this pair of datasets.

VI. LIMITATIONS AND FUTURE WORK

On the experimental design side, Rabbani’s results rest on a single fixed 70/15/15 split rather than a cross-validated one; a classic k -fold cross-validation – a plain random split into k folds, without stratification, since no grouping variable analogous to HemoSet’s subjects is available – would provide a variance estimate comparable to HemoSet’s 5-fold results, together with a closer look at why one particular

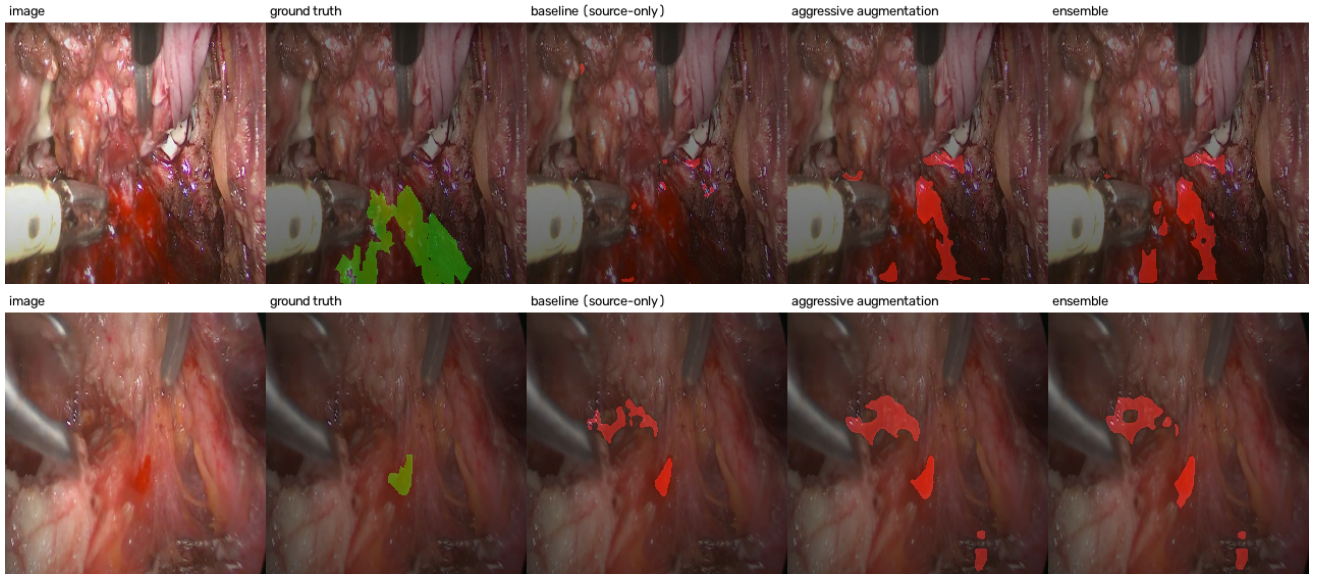


Fig. 3. HemoSet→Rabbani, two examples: image, ground truth, baseline, aggressive augmentation, ensemble. The top example shows the baseline’s characteristic under-segmentation (low recall), progressively recovered by aggressive augmentation and the ensemble.

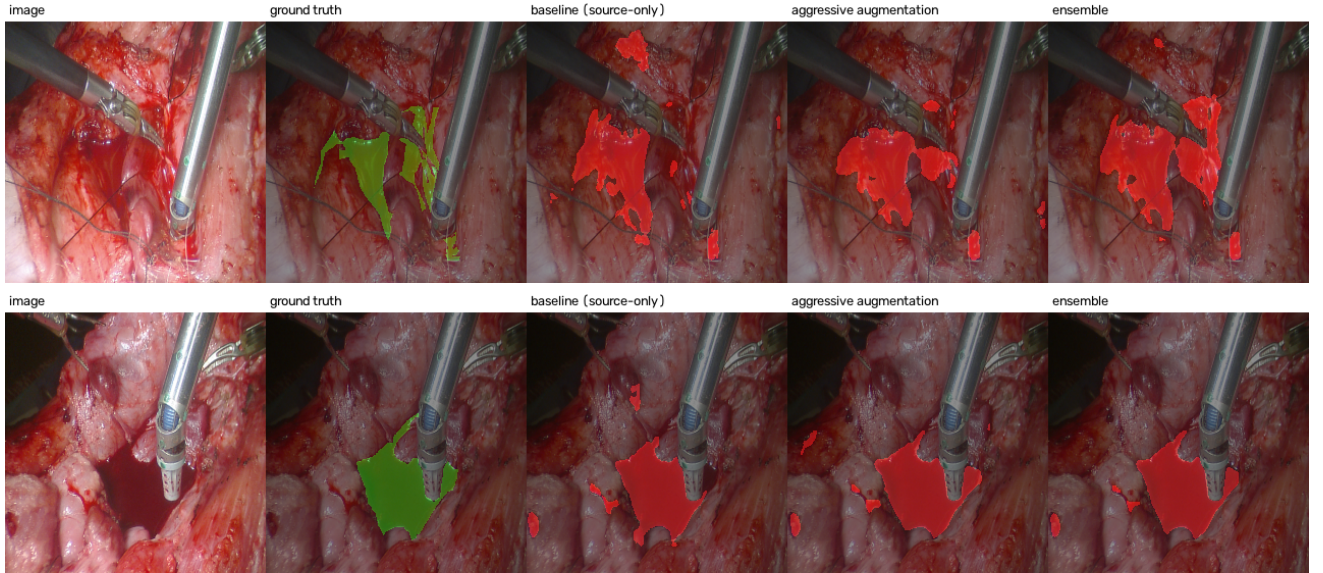


Fig. 4. Rabbani→HemoSet, two examples: image, ground truth, baseline, aggressive augmentation, ensemble. The top example shows the characteristic over-segmentation (false positives) of this direction; the bottom example shows a cleaner case with only minor residual false positives.

HemoSet subject is consistently harder to segment than the rest. On the methodological side, two refinements to the strategies already tested represent natural next steps: applying appearance/domain adaptation with probability < 1 instead of always-on, to check whether this alone explains the in-domain cost observed in Section IV-B, and combining it with aggressive augmentation to see whether the two effects compound rather than compete.

Two directions that were not explored trade simplicity for potentially larger gains at the cost of additional complexity: self-training with pseudo-labels on the target domain, and the style-content disentanglement or Transformer-backbone

approaches discussed in Sections II and V-B. Finally, moving toward deployment, the ensemble could be extended across folds as well as architectures, and the recommended configuration should be re-evaluated on real TRAMIS data once available, keeping in mind the domain-balancing lesson from the joint-training experiments, since mixing external datasets of very different sizes is not automatically beneficial without it.

VII. CONCLUSIONS

This work systematically quantified the asymmetric domain gap between HemoSet and Rabbani, and compared six families

of strategies for increasing the cross-dataset robustness of surgical blood segmentation models. Aggressive data augmentation proved to be the most effective and lowest-cost strategy among those tested; combined with an ensemble of three architectures, it produces the best overall result, with no additional training beyond what augmentation already requires. Three further extensions did not improve on this result, but their analysis supports the hypothesis that the key factor is preserving the model’s exposure to the “clean” distribution of its own domain during training, rather than imposing invariance through signals or normalizations computed directly on the data.

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